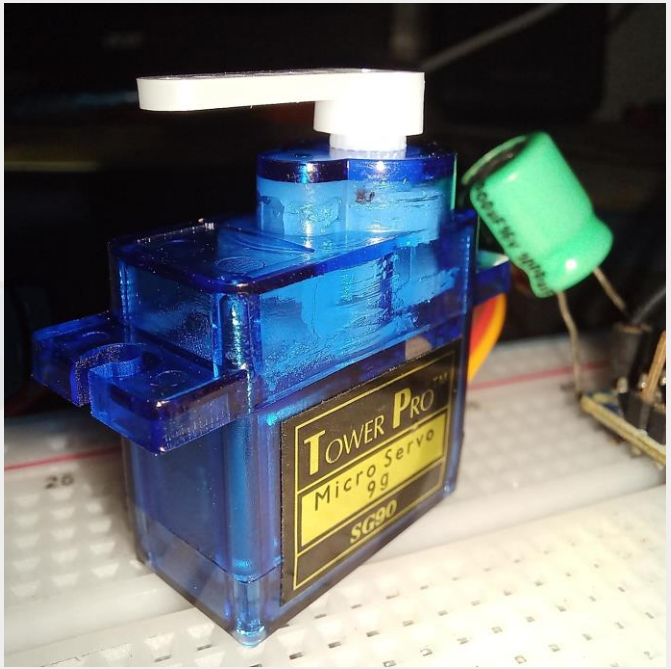
Servo Motor

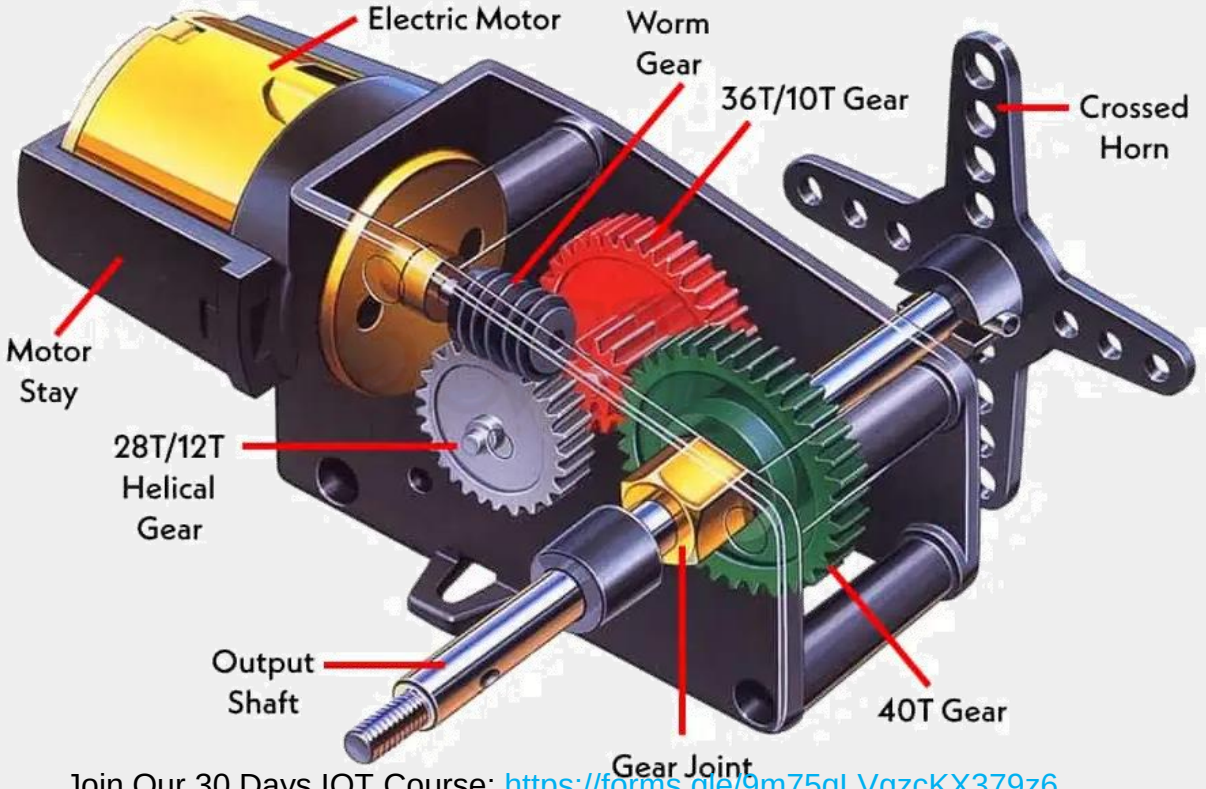
Required Features:

- Converts electrical energy to mechanical force.
- Precise movement possible
- Range: 0-180 degree
- 1 Digital wire input
- High endurance
- Low cost



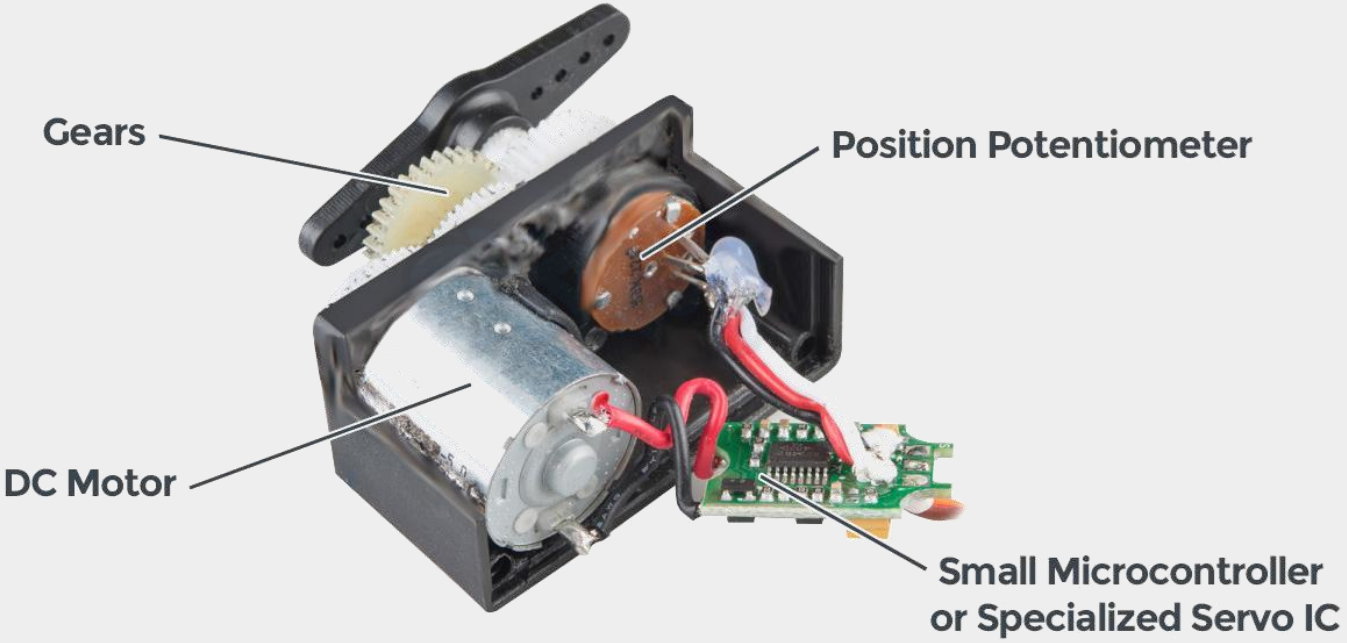
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Internal Architecture



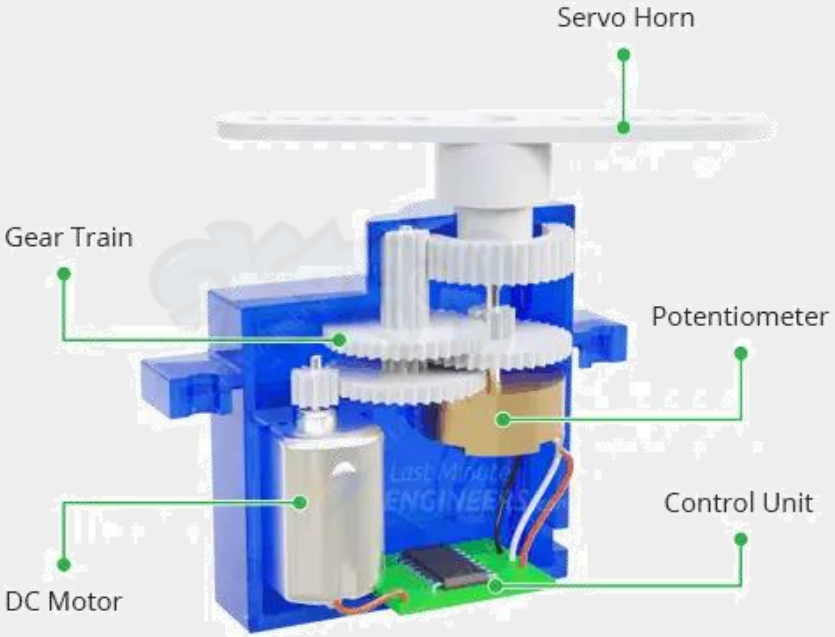
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Internal Architecture



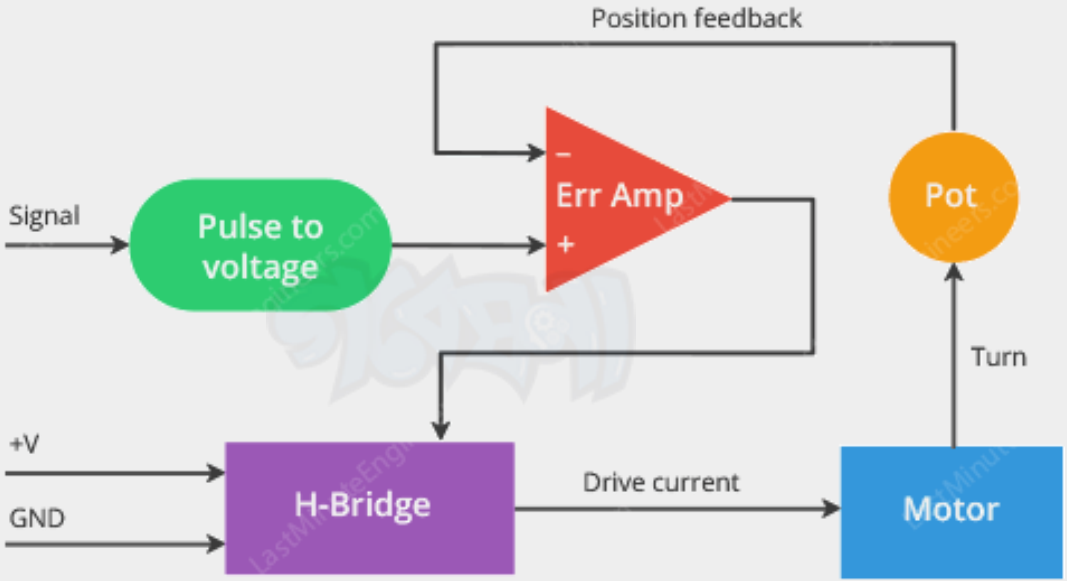
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Internal Architecture



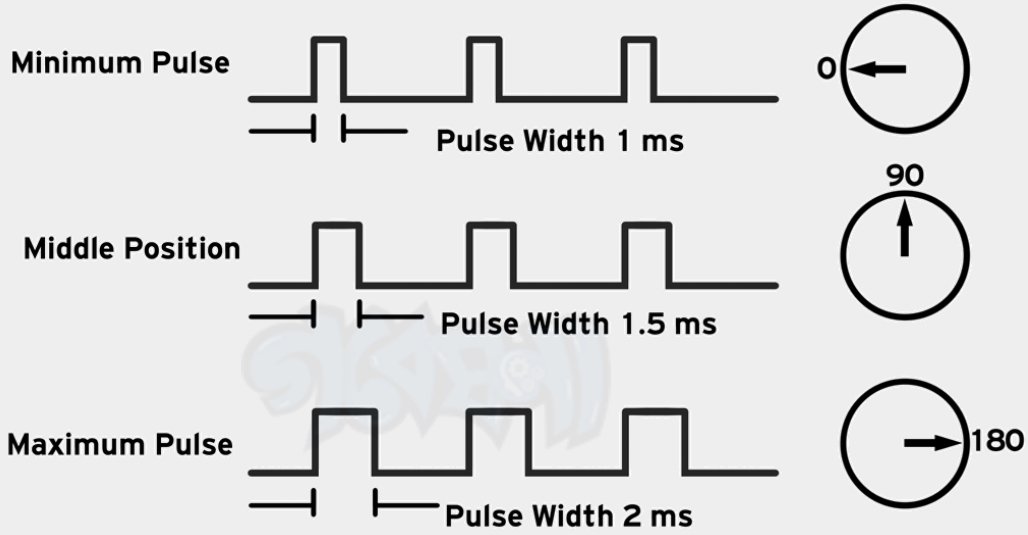
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Block Diagram



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Timing Diagram



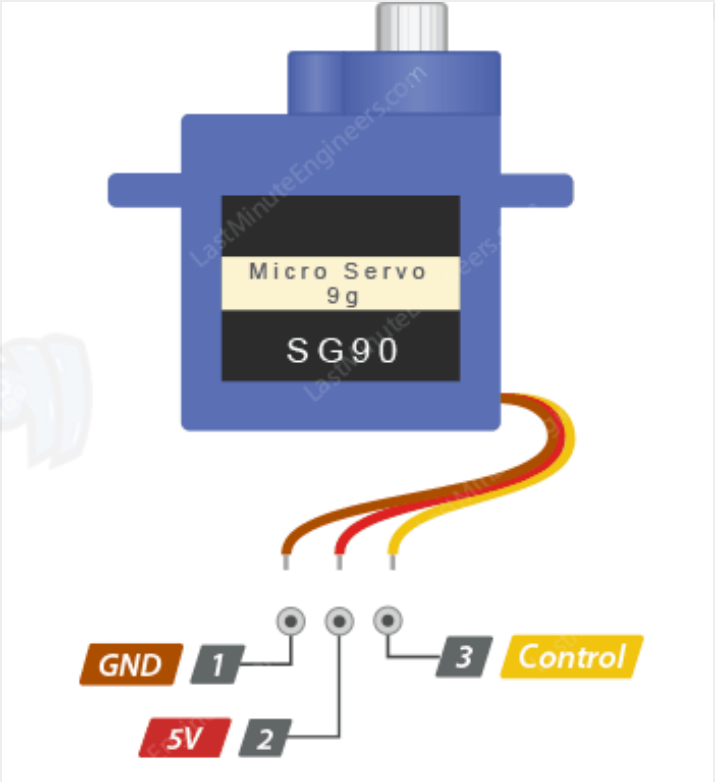


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Servo Pinout

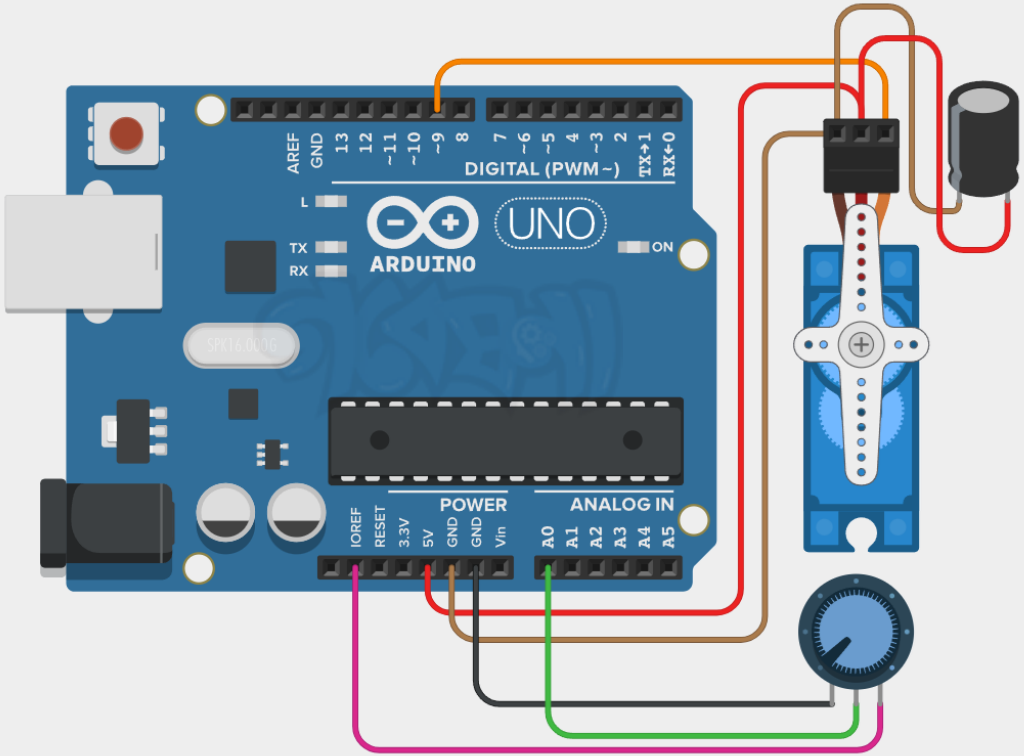
Specifications:

- Power rating: DC 5v 250mA
- Yellow: Control pin
- Red: Power (5v)
- Brown: Gnd



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Sample Circuit



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Assignment

Write a code for **Automatic Door lock system using PIR & Servo.**

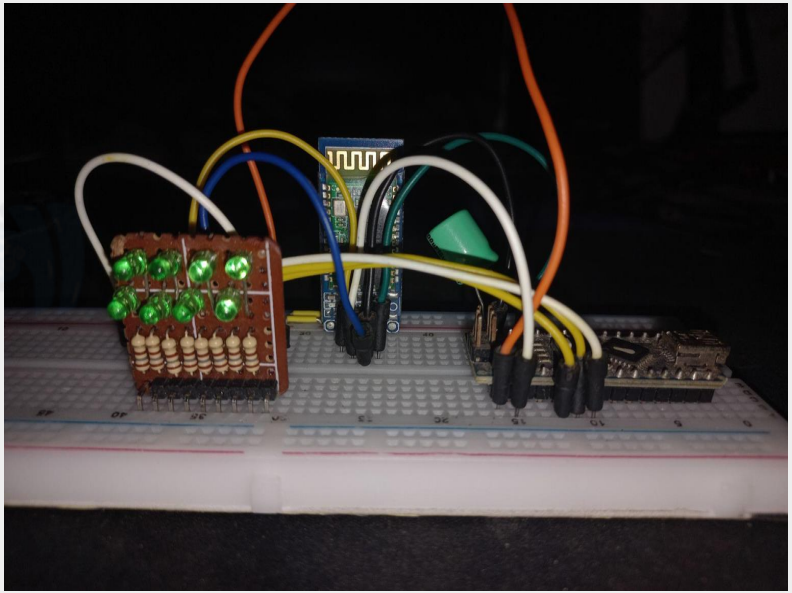
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Interface with Bluetooth

Required Features:

- Send/receive any command over Bluetooth.
- Hassle free operation.
- Traditional Serial communication.
- 2 pin used: RX, TX.
- Operated with Android App.

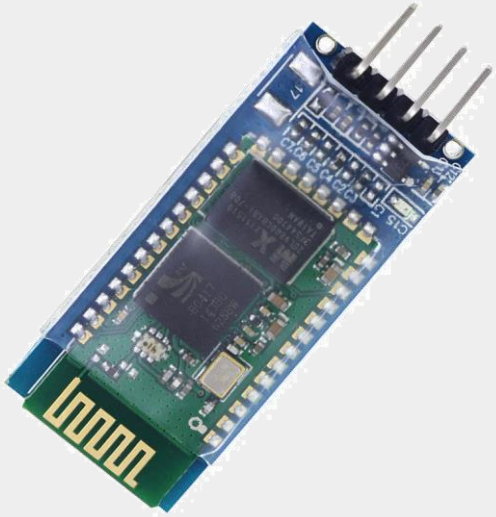


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Bluetooth Module

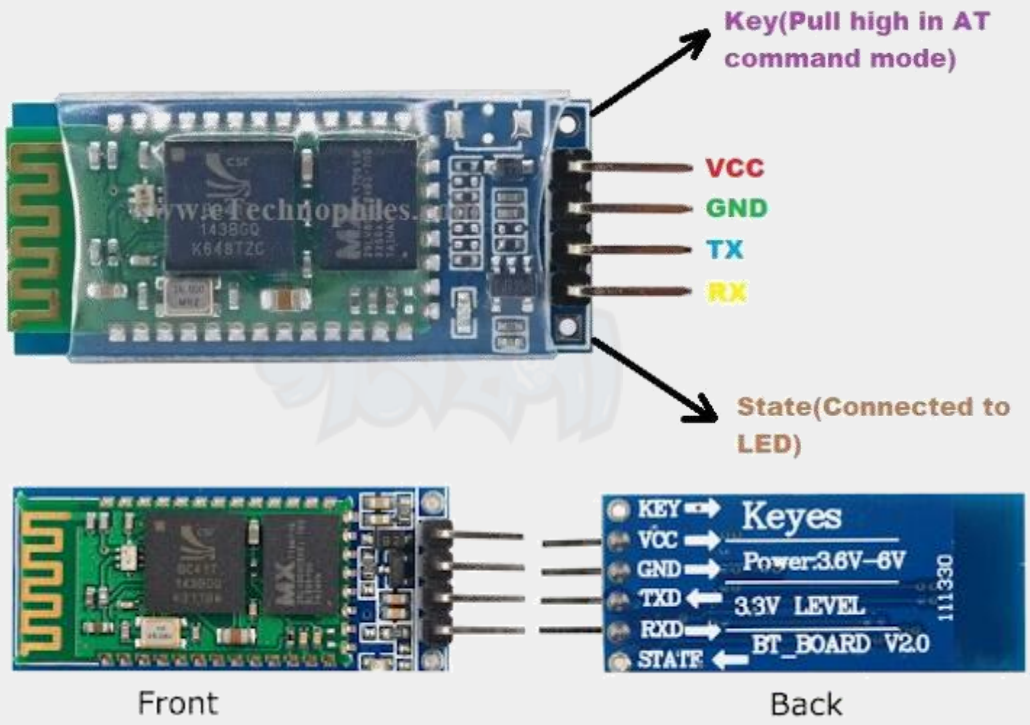
Specifications:

- Model: HC 06
- Power rating: DC 3.3V-6V
- BT Profile: Serial
- RX/TX: 3.3V
- Max Range: 10 meter



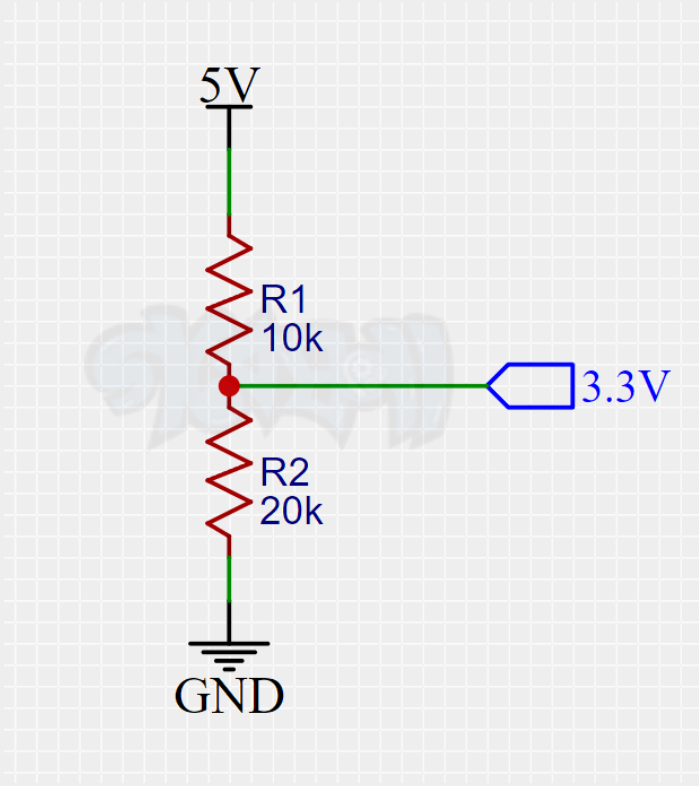
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HC 06 Pinout



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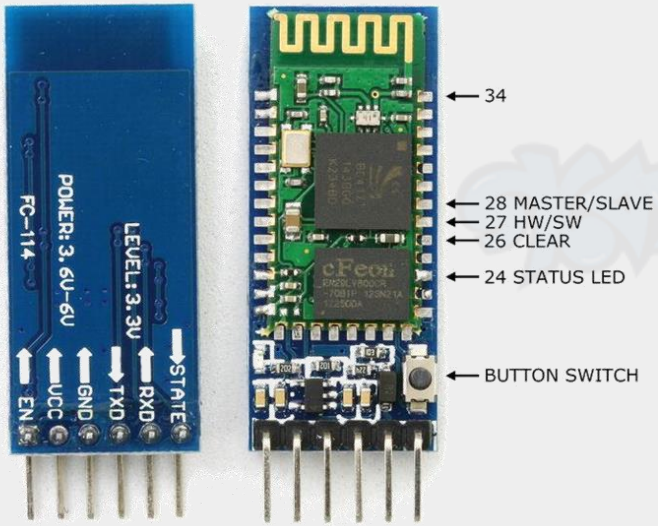
Level Shift



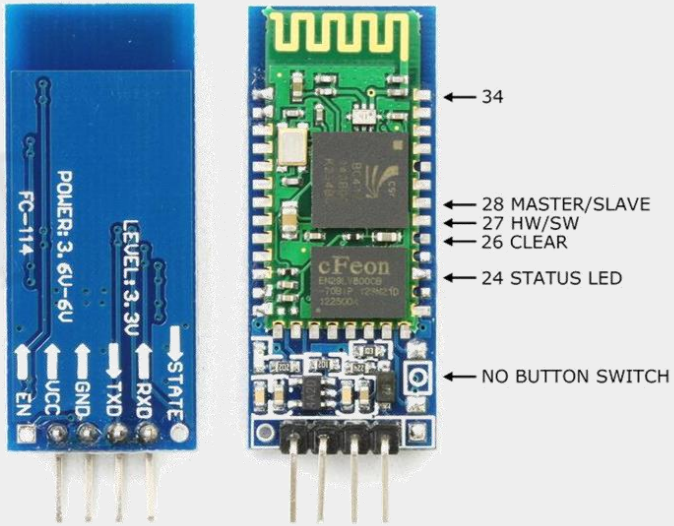
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Model Compare

HC-05 FC-114



HC-06 FC-114



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Android App

Arduino bluetooth controller

Giumig development apps

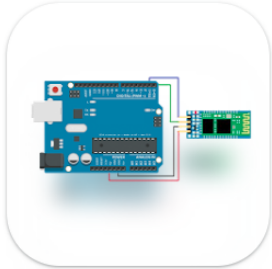
In-app purchases

4.2★ 3.23K reviews | 1M+ Downloads | 3+ Rated for 3+

Install | Share | Add to wishlist

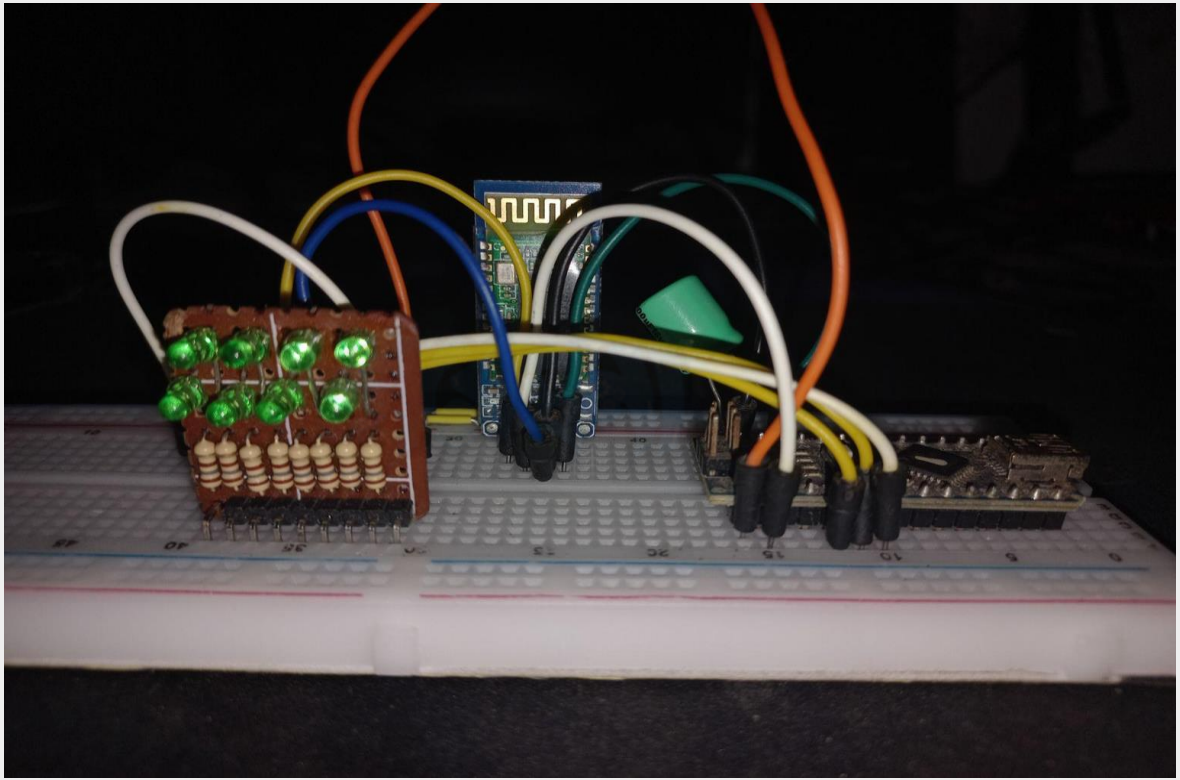
This app is available for all of your devices

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Sample Circuit



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Assignment

Write a code **to transmit a sensor reading from arduino to android app over bluetooth.**

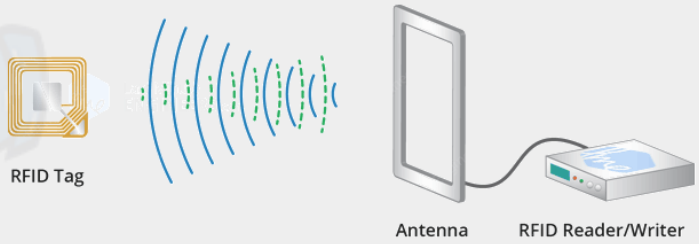
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Interface with RFID

What is RFID?

- Radio Frequency Identification
- Contactless Communication.
- 1 Tag, 1 Reader with an antenna
- 2 protocols: Low speed 125KHz,
  High Speed: 13.56 Mhz
- Quick response time.



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RFID Reader Module

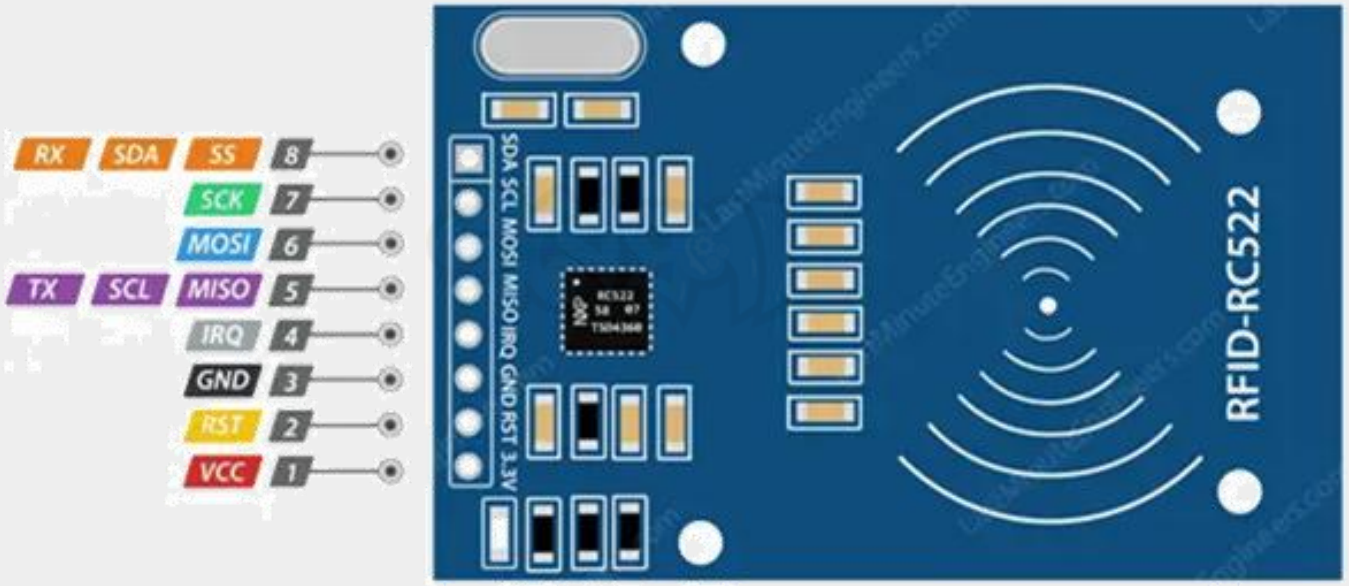
Specifications:

- Model: RFID-RC522
- Manufacturer: NXP Semiconductor
- Power rating: DC 3.3V 26mA Max
- Reading range: 5CM
- Frequency: 13.56 MHz ISM Band
- Protocol: SPI / I2C / UART



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Pinout

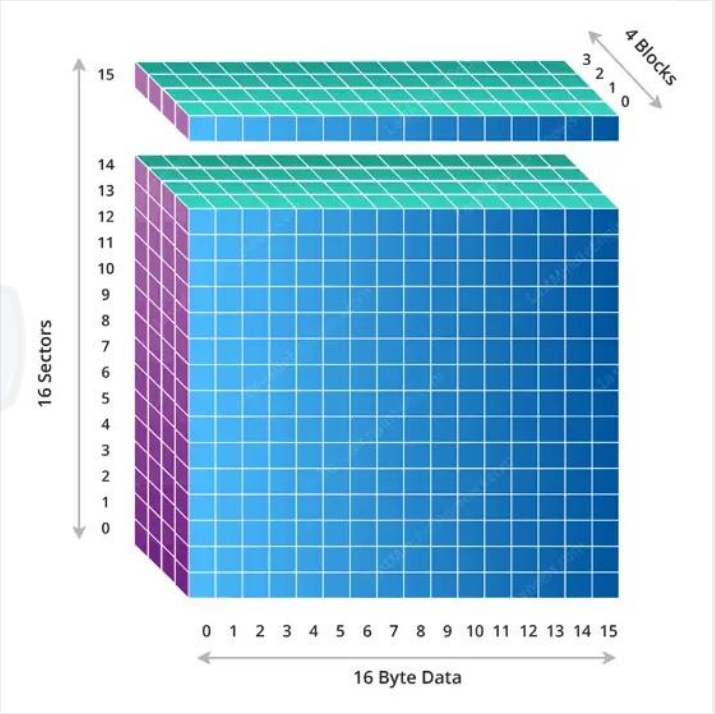


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Memory Structure

1k Memory Layout:

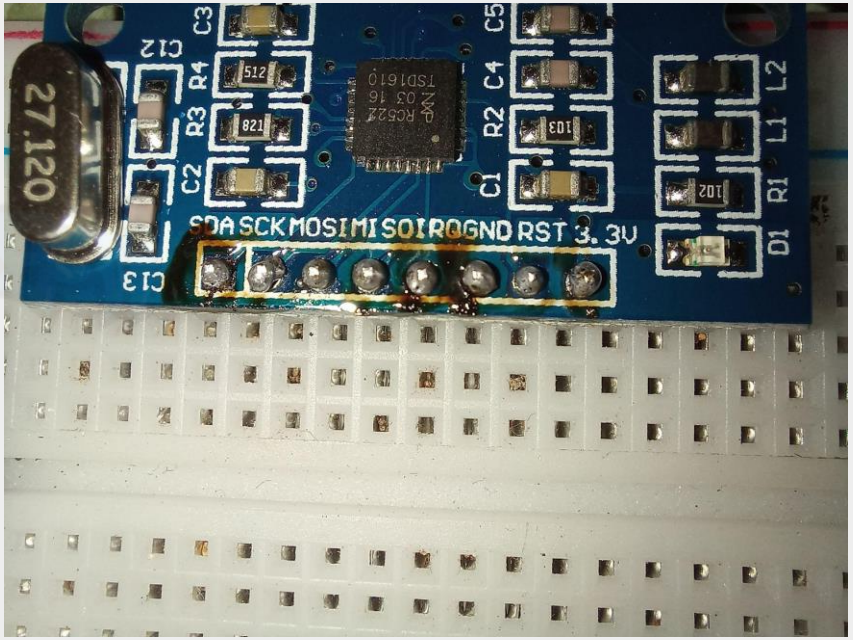
- Total Sectors: 16 (0-15)
- Each sector divided by 4 Blocks.
- Each block contains 16 bytes of data
- 16 x 4 x 16 = 1024 bytes = 1K memory
- Block 0, Sector #0 is the Manufacturer Block. Contains 4 bytes of UUID



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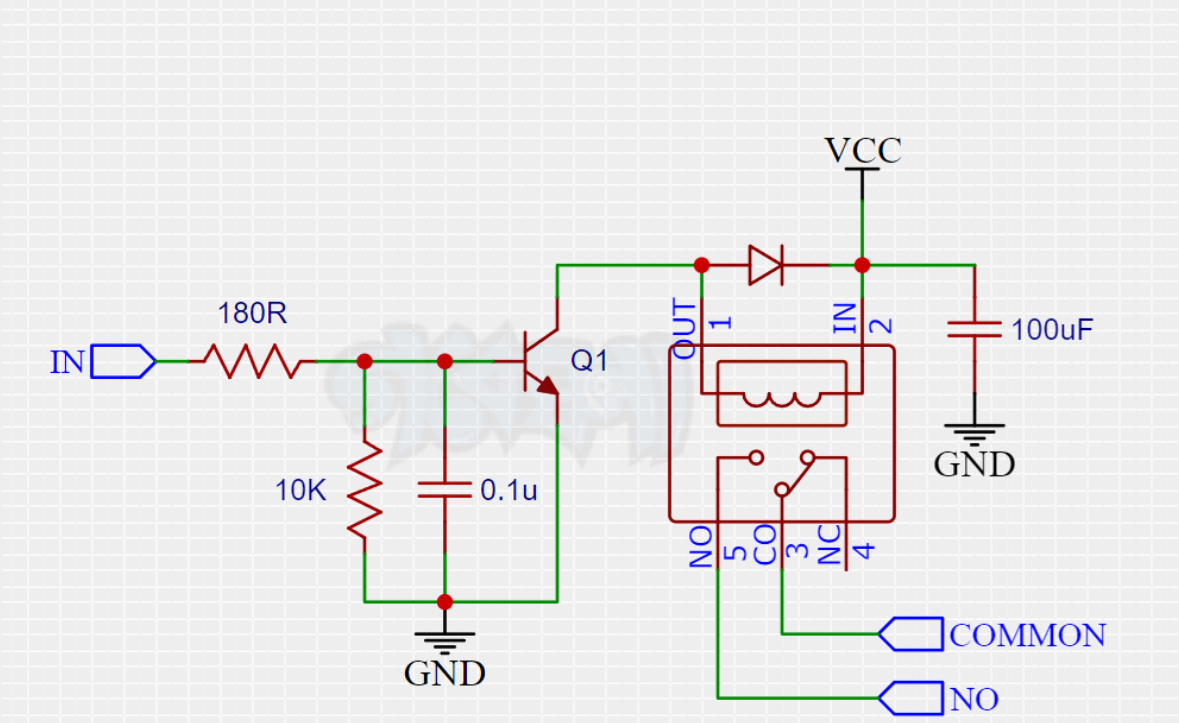
Interface with Arduino

RFID pin	Arduino Pin
VCC	3.3V
GND	GND
RST (Reset)	9
MISO	12
MOSI	11
SCK	13
SS/SDA	10



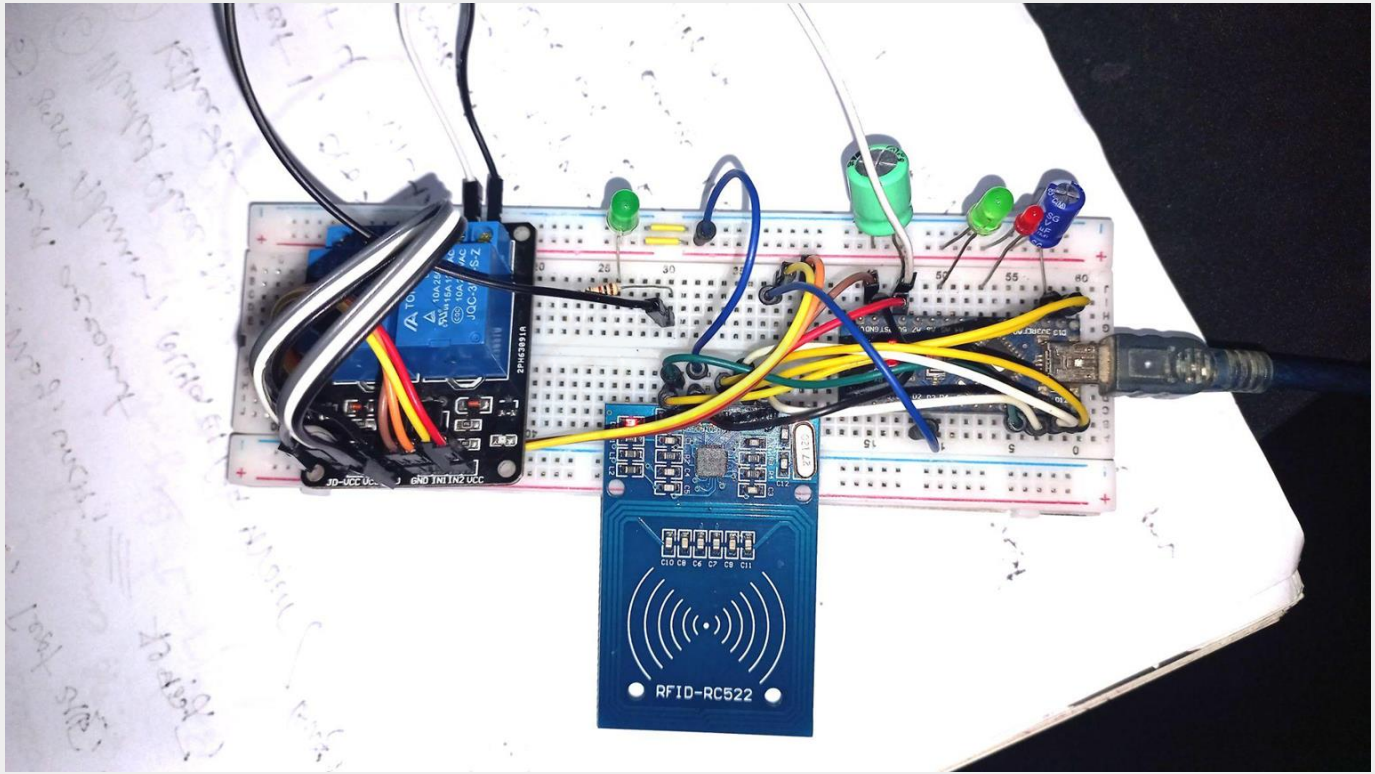
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Relay Circuit



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Sample Circuit



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Assignment

Write a code for arduino **to emulate the door lock with RFID authentication.**

Store the card info inside an EEPROM chip. (Optional)

Include a button to save the card info into the EEPROM. (Optional)

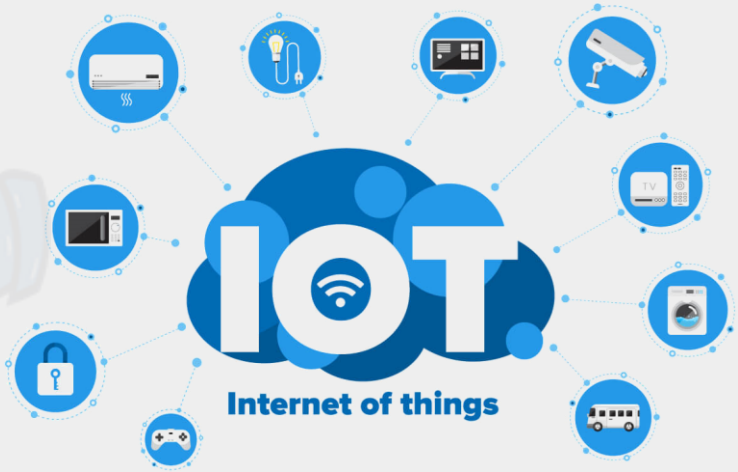
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IoT in Arduino

What is IoT?

- Remotely control any device through the internet.
- Automation and monitoring will be much easier.
- Everything is centrally connected



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ESP8266 Microcontroller

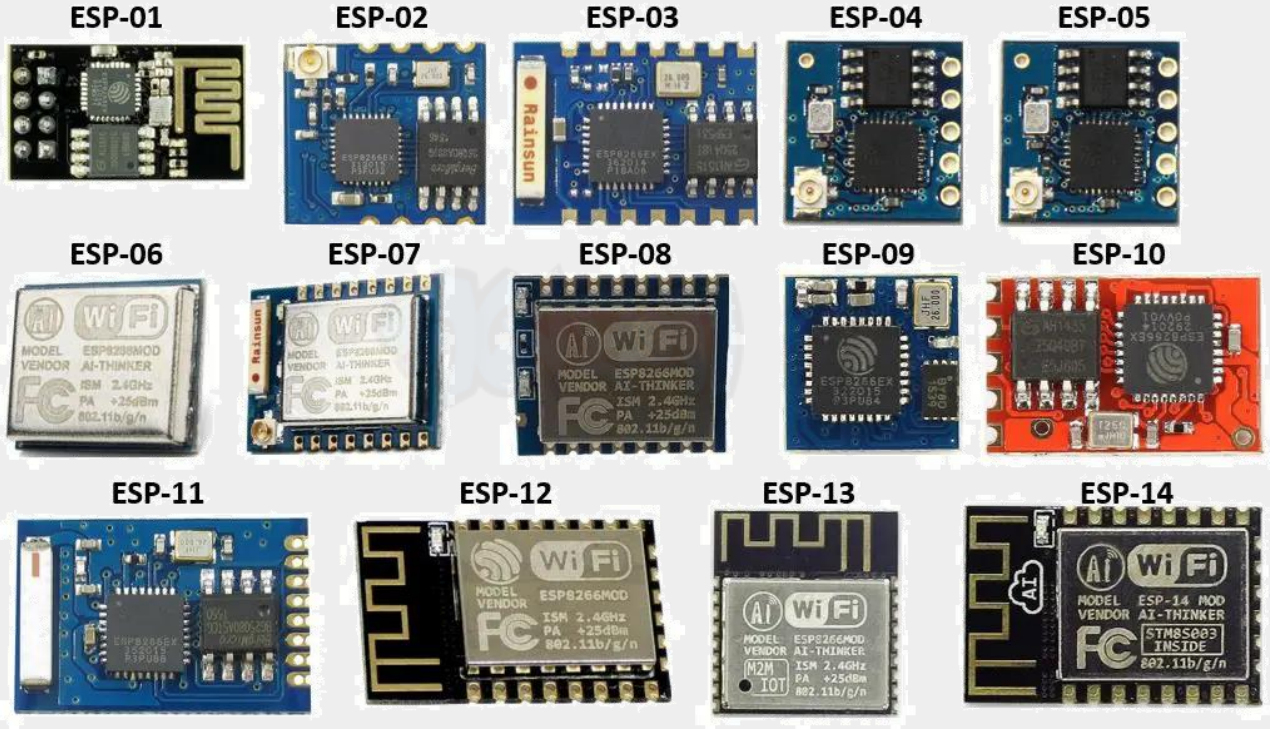
Specifications:

- Manufacturer: Espressif Systems
- Tensilica Xtensa® 32-bit LX106 RISC microprocessor 80-160 MHz clock speed.
- 802.11b/g/n WiFi interface
- Interfaces: ADCs, UART, SPI, I2C, PWM.
- 3.3v Operation, low cost.



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Available Models

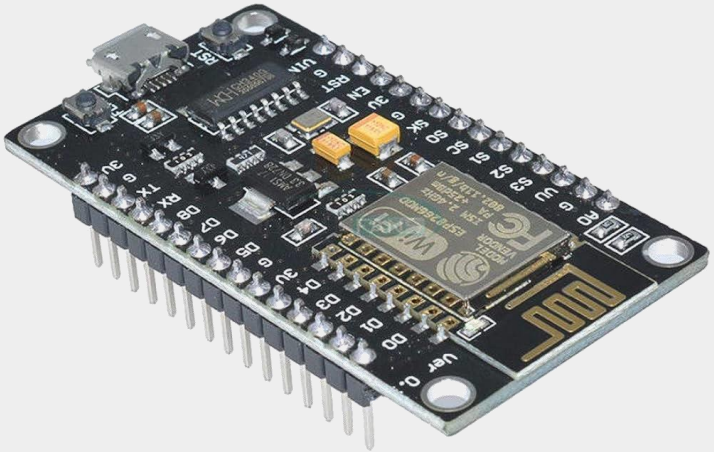


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NodeMCU Board

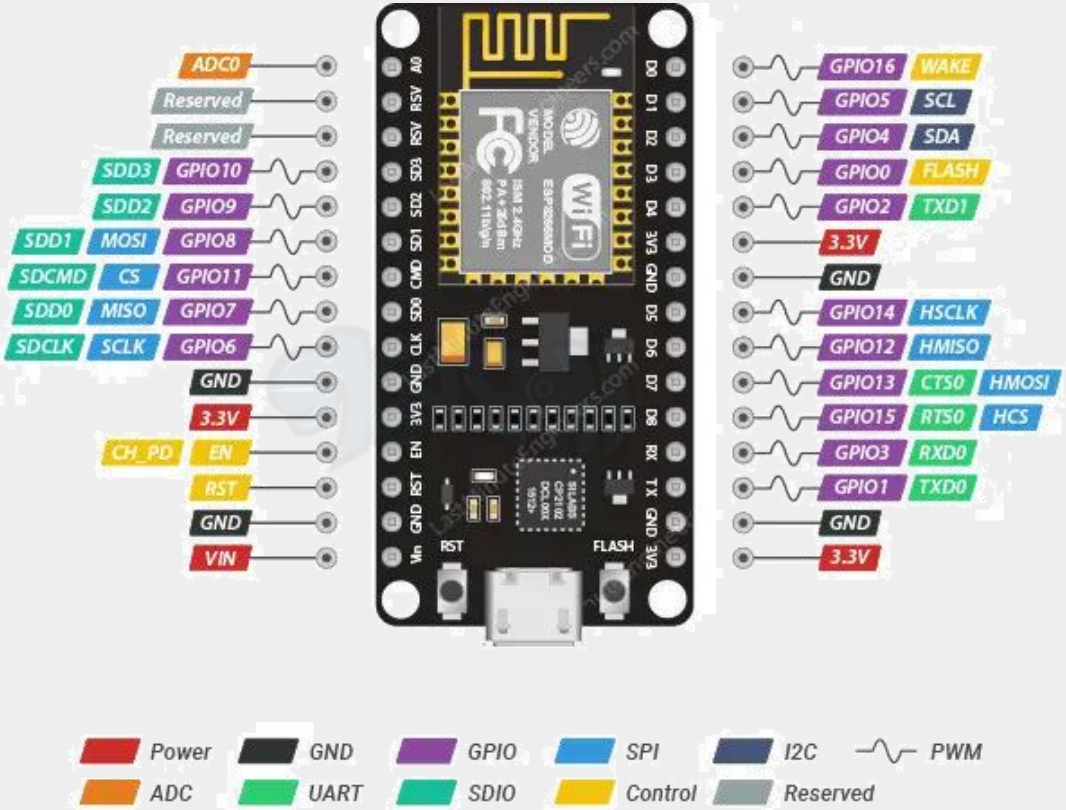
Specifications:

- Model: ESP8266 NodeMCU
- Interface: UART, SD, I2C, SPI
- 1 channel 10Bit ADC
- 1 on board built-in LED.
- 11 GPIO pins, all are PWM capable.
- 128KB internal RAM, 4MB external Flash
- 3.3v Low power operation.
- Wi-Fi interface (Soft AP, Station mode)



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NodeMCU Pinout



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ThingSpeak

About ThingSpeak:

- Web Server that connects different IoT devices over the internet.
- Maximum 8 values can be read/write
- 15-20s refresh interval
- Free to use (8200msg/day)
- Web login required.

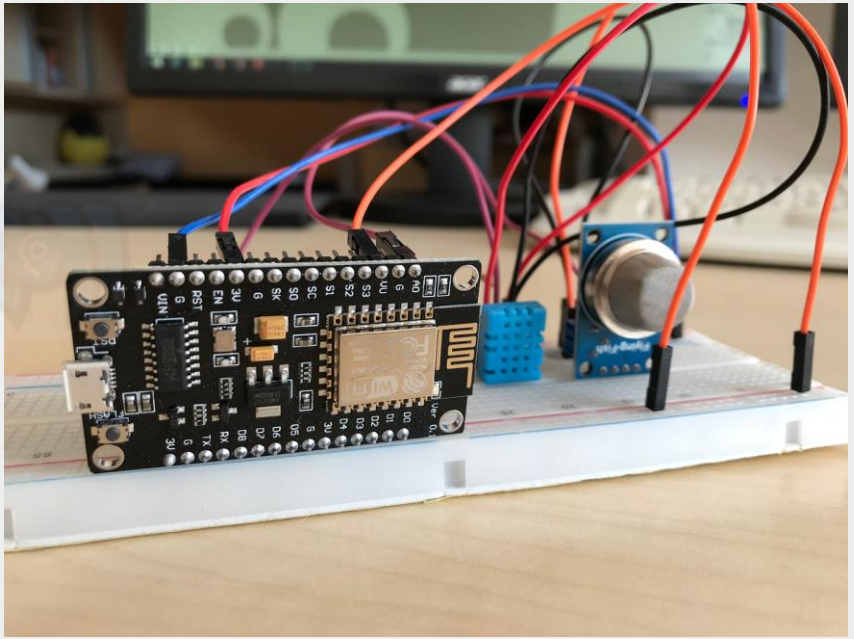


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Demo Project

Requirements:

- Monitor the ambient temperature through the internet.
- Live data monitoring.
- Simple operation.
- Easy monitoring from web.



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Assignment

Write a code for arduino **to monitor ambient temperature and humidity with ESP8266 and send those data to ThingSpeak server.**

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Memory

Memory, why?

- Hold the value of local/global variables in S-RAM.
- Save Firmware in Flash memory
- Save configuration info from user.
- Save Large data for logging purpose in SD card.

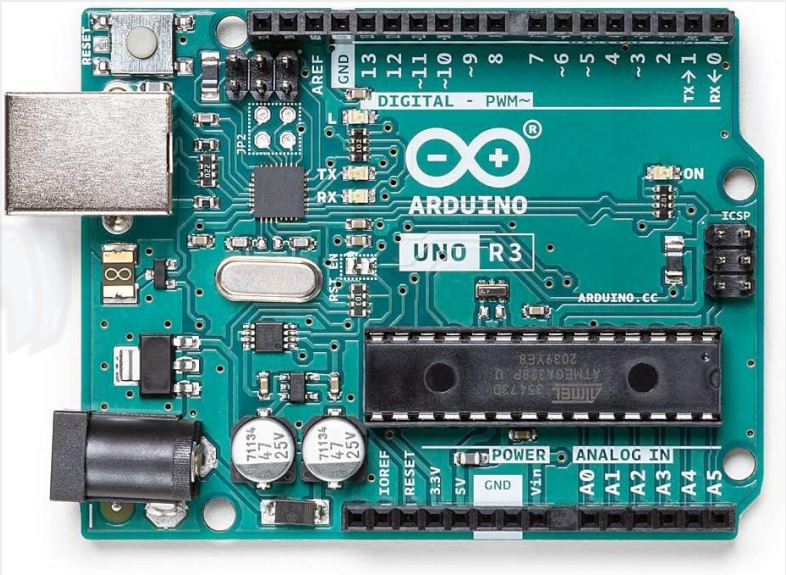


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Memory of Arduino

Memories in ATMEGA328P:

- SRAM: 2KB
- Flash: 32KB
- EEPROM: 1KB (1024 Bytes)
- 1M write cycle for EEPROM



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Internal EEPROM Access

EEPROM Library: `#include <EEPROM.h>`

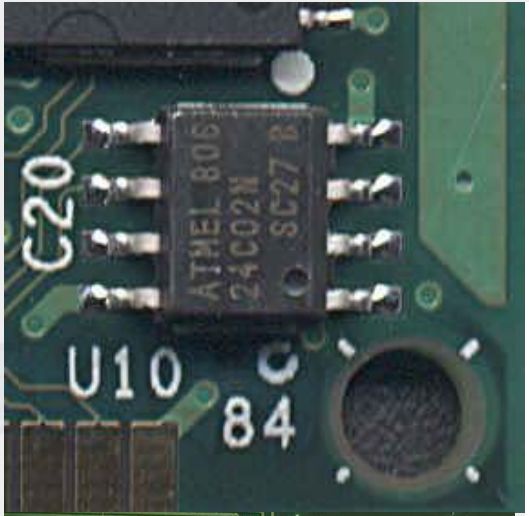
- Read 1 byte of data: `EEPROM.read(address)`
- Write 1 byte of data: `EEPROM.update(address, value)`
- Read any datatype: `EEPROM.get(address, data)`
- Write any datatype: `EEPROM.put(address, data)`

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EEPROM Usage

- TV, Refrigerator
- Air Conditioner
- Automobile industry
- Industrial automation
- Computer peripherals
- Gaming Consoles

  and so on...







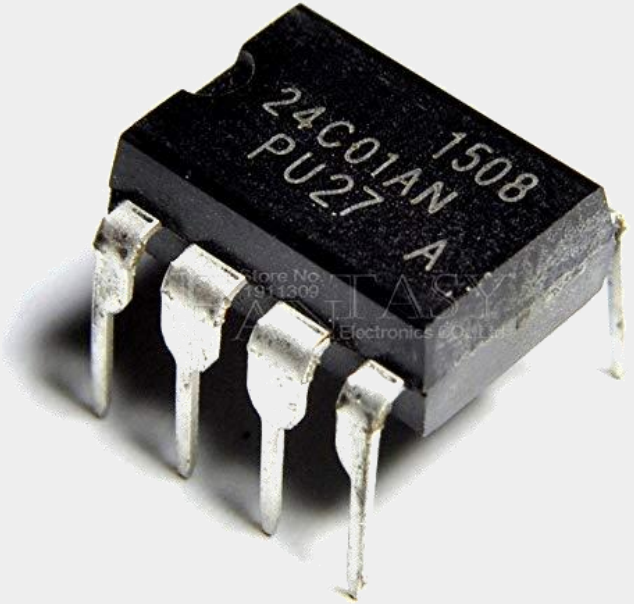


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External EEPROM

Specifications:

- Model: AT24Cxx (01-512)
- Storage Capacity: 128 x 8
  = 1Kbits
  = 128 bytes
- I2C interface
- 1.8-5v operation
- Endurance: 1 Million Write Cycles
- 3 pins for addressing
- Device address 1010XXX.

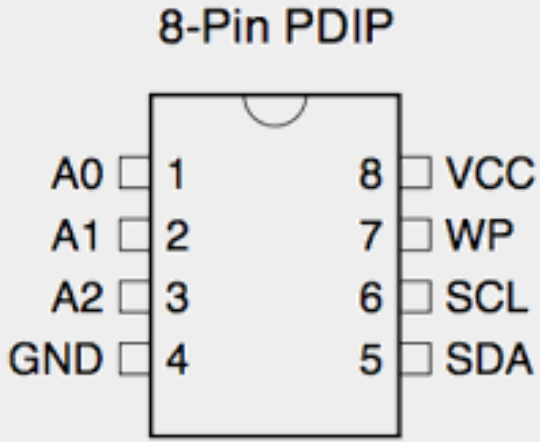


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PinOut

AT24CXX Pins:

- **A0-A2: Device Address Selection**
- **WP: Write Protect.** Normal (GND)
- **SCL: i2c Serial Clock**
- **SDA: i2c Serial Data**
- **VCC: 1.8-5V**
- **GND: Ground**

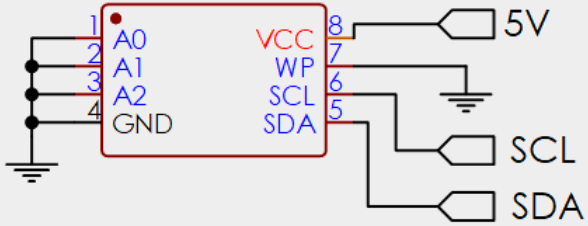


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Sample Circuit

AT24C01 Pins:

- A0-A2: Arduino GND
- WP: Arduino GND
- SCL: To Arduino A5 (SCL)
- SDA: To Arduino A4 (SDA)
- VCC: 5V
- GND: Ground
- Device Address: **1010 000**



External EEPROM Library:
https://github.com/manzarehassin/AT24C16

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